9

Long-term management of direct restorations

Chapter contents

9.1	Introduction	167
9.2	Restoration failure	168
9.3	Tooth failure	172
9.4	Monitoring the patient/course of the disease	174
9.5	Managing the failing tooth-restoration complex: the '5 Rs'	175
9.6	Answers to self-test questions	179

9.1 Introduction

As has been emphasized throughout this book, minimum intervention oral/dental care involves more than just the minimally invasive operative treatment of the consequences of dental disease. It involves identifying and predicting disease patterns, and concerns the control/prevention of disease by modifying aetiological factors and *reassessing* the adherence to changes in patient behaviours, attitudes, and responsibility. *Monitoring* the oral cavity and restored dentition ensures that the treatment undertaken, and subsequently improved oral health, is *maintained*. This should be accomplished through individualized strategic *recall* regimes. Restorations need to be *reviewed* regularly and occasionally *refurbished, resealed, repaired*, or *replaced* (see Figures 9.1, 9.2, and 9.3, and Section 9.5). Therefore periodic recall appointments, once an episode of treatment has been completed, are just as important as the treatment itself. It is critical that the patient understands the importance of these recall consultations as part of the ongoing care that is being offered to help to maintain their oral health.

Three aspects of dental care need to be assessed at recall visits:

- the overall state of the patient's oral and dental health (review)
- the individual patient's longer-term response/adherence to previous preventive advice and/or treatment, in moderating any aetiological factors that could cause future dental disease (reassessment)
- the status and quality of the restorations present (monitoring and maintenance).



Figure 9.1 Corrosion products from a distal amalgam in the UR3 causing coronal discoloration which the patient complained about—an example of aesthetic restoration failure.

Q9.1: What restorative material should be used to replace the amalgam shown in Figure 9.1?



Figure 9.2 An aged, discoloured mesial resin composite restoration on UL2. This caused aesthetic concerns for the patient, who requested its replacement.

Q9.2: What would be the difficulty faced by the dentist when replacing the Class III restoration shown in Figure 9.2?

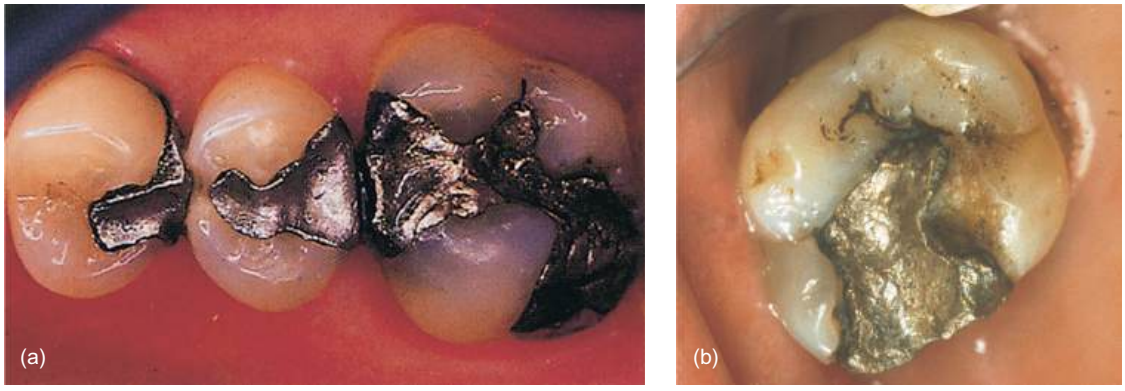


Figure 9.3 (**a**) A series of old, ditched, and partly corroded amalgam restorations, that were observed on clinical examination to be cleansable and not causing direct problems to the patient. Although the margins and occlusal surfaces are not perfect, should these amalgams be classified as failed restorations? (**b**) A ditched amalgam UL7 with a marginal defect large enough to accommodate the tip of a periodontal probe, accumulating plaque. A combination of refurbishment with repair or replacement in this case would facilitate plaque control.

9.2 Restoration failure

The potential causes of restoration failure have been identified and outlined in Table 9.1. It is important to appreciate that the causes of restoration and tooth failure (see Table 9.2) are often multifactorial in nature. Indeed, as the causes of both tooth and restoration failure are inextricably linked, it is wise to consider them together, as a *tooth–restoration complex*.

9.2.1 Aetiology

The multifactorial aetiology of restoration failure is often due to manifestations of inherent long-term weaknesses in the mechanical properties of different restorative materials (e.g. poor edge strength, wear, compressive strength, water absorption, etc.) and/or problems with the technical application of the restorative material for the chosen clinical situation (i.e. incorrect choice of material and poor placement technique).

9.2.2 Choice of restorative material

The chemistry and physical properties of the different direct, plastic restorative dental materials at a dentist's disposal have been discussed in Chapter 7. Restorative material longevity will be affected by:

Table 9.1 Causes of restoration failure and the criteria used to assess this failure, with clinical comments; it is vital to appreciate that multiple aetiologies of restoration failure are common and must be considered alongside the causes of tooth failure—the tooth–restoration complex

Restoration failure criteria	Causes and clinical comments
Colour match (aesthetics)	• Important to get the patient's views, especially in the anterior aesthetic zone—they may or may not be concerned. Ability to manage patient expectations is important at the outset of any treatment offered • Underlying discoloration from stained dentine • Superficial discoloration from marginal/surface staining • Underlying discoloration from corrosion products (e.g. amalgam; see Figure 9.1) • Aged tooth-coloured restorative materials become stained and discoloured due to water/food stain (tannin) absorption leading to a gradual change in optical properties (see Figure 9.2)
Margin integrity	• Loss of margin integrity (*allowing plaque stagnation*) caused by: ◦ long-term creep/corrosion/ditching of amalgams (see Figure 9.3) ◦ margin shrinkage of resin composites/bonding agent (see Figure 9.4) ◦ margin dissolution/shrinkage (on dessication) of glass ionomer cements ◦ margin chipping under occlusal loading due to poor restoration edge strength ◦ presence of ledges/overhangs, poor contour of margins (see Figure 9.5) • If the patient can keep the failed margin plaque- and recurrent caries-free and it is not of aesthetic/functional concern, this partial loss of integrity may not be a strong indicator to repair/replace the restoration
Margin discoloration	• Micro-/macro-defects at the tooth–restoration interface will permit exogenous stain (e.g. food stains) to penetrate along the outer perimeter of the restoration as well as towards the pulp • Poor aesthetics (see Figure 9.6) • An indication of margin integrity failure? • Not necessarily an indication of secondary/recurrent caries/caries associated with restorations/sealants (CARS)
Loss of bulk integrity	• Restorations may be bulk fractured/partially or completely lost due to: ◦ heavy occlusal loading—inadequate occlusal assessment before restoring the tooth (see Figure 9.7) ◦ poor cavity design leading to weakened, thin-section restorations (especially for amalgams; see Figure 9.8) ◦ poor bonding technique/contamination leading to an adhesive bond failure and lack of retention ◦ inadequate placement procedures (condensation technique/curing) causing intrinsic material structural weaknesses (e.g. voids, 'soggy bottom' in resin composites) • Patients will often complain of a 'hole in the tooth' where food debris is trapped—↑ caries risk (see Figure 9.9) • Bulk loss of restoration or occlusal wear may affect the bite/occlusal scheme (see Figures 9.10 and 9.11)



Figure 9.4 Occlusal resin composites on UL6. Note the poor aesthetics and margin staining/breakdown. This restoration would be clinically easier to replace than to repair, due to its small size, in a high caries risk patient with continuing poor oral hygiene, as plaque stagnation will be a concern at the defective restoration margin.

- the differences in physical properties of each material (e.g. bulk amalgam has the greatest strength and resistance to wear, glass ionomer cement (GIC) has limited long-term wear resistance, resin composites exhibit volumetric shrinkage)
- the occlusal loading placed on individual restorations (which might cause some materials to wear/fatigue/fracture more quickly than others)
- the rapid development of dental material science—as the mechanical properties and placement techniques are being refined continuously, so the longevity of new materials will surely improve



Figure 9.5 A bitewing radiograph of an overhanging (positive ledge) radio-opaque amalgam restoration on the mesial aspect of UL6. Due to plaque stagnation, mesial alveolar bone loss has occurred, along with radiographic evidence of caries (radiolucency beneath the overhang—care is needed to distinguish this from radiographic cervical burnout).

Q9.3: Look at Figure 9.5. What has caused this positive ledge to occur?



Figure 9.6 Marginal staining evident on the resin composite restorations, distal UL1, mesial and distal UL2, and buccal cervical UL3. The patient complained of poor aesthetics. Note the plaque accumulation at the gingival margins of UL2 and UL3, posing a caries risk in these areas.

Q9.4: Look at Figure 9.6. What periodontal condition has been caused by the accumulated plaque, and what is its relevance when replacing the restorations?

- the simplicity of the clinical handling characteristics of the material (the easier it is to manipulate by the nurse/dentist, and the fewer stages required for its placement, the less opportunity there is for iatrogenic weaknesses to be introduced)
- in addition to the previous point, the clinical skill of the operator. It is these last two points that are most critical in affecting restoration longevity in modern conservative dentistry.



Figure 9.7 The resin composite restoration on LR6 has fractured at the distal marginal ridge due to a heavy occlusal contact with the opposing cusp.



Figure 9.8 A partially fractured occlusal amalgam restoration in LL6.

Q9.5: What factors have resulted in the fracture shown in Figure 9.8?

Clinical research studies and statistical meta-analyses of past clinical data have attempted to answer objectively the question of how long restorations should last (see QR code image 9.1). However, due to the numerous uncontrollable variables already mentioned (primarily the operator and the patient), obtaining a precise figure is impossible and arguably irrelevant! Often quoted *average* age ranges of restorations at the time of replacement are as follows:

- amalgam restorations: 10–15 years
- resin composite restorations: 5–8 years
- GIC restorations: 3–5 years.

QR code 9.1 Scan this code with your mobile device to access MI Compendium, an online resource of collected systematic reviews of MI dentistry techniques and materials.

www.mi-compendium.org



Figure 9.9 Fractured amalgam restorations UR7 mesial and UR4 distal, which need repair due to plaque stagnation and recurrent caries (CARS).



Figure 9.10 Occlusal wear on resin composite restoration LR6.

Q9.6: Look at Figure 9.10. Does this finding necessitate the replacement of this restoration?



Figure 9.11 (**a**) Posterior occlusal restoration on LL6 at time of placement (blue marks on teeth are made by articulating paper to check the occlusion). (**b**) The same restoration 5 years after placement, showing early signs of occlusal wear. (**c**) The same restoration 9 years after placement, with further loss of occlusal definition. The margins were intact and there were no symptoms from this tooth, but the aesthetics were a little compromised. This does not constitute a restoration failure, and the restoration can continue to be monitored.

However, this does not mean that all GICs, for example, will fail catastrophically after 5 years, or that all amalgams will last for 15 years without any problems. The hotly debated issue regarding the assessment criteria for designating failure (see Section 9.2.3) rears its head when considering the previously cited longevity figures. It must be appreciated that the weighted clinical importance of some of the criteria for failure assessment (outlined in Table 9.1) will be dependent on the restorative material being assessed. For example, when considering aesthetics as an assessment of restoration failure, a black, tarnished amalgam with ditched, corroded margins on a premolar tooth may be considered still functional (as it probably would be detrimental to the remaining tooth structure to replace it), whereas a mildly partially stained resin composite may be considered seriously for repair/replacement (as the repair procedure is perceived to be relatively simple and non-destructive by dentists). Which restoration has actually 'failed'? The most useful answer to the lead question in this section is one given by an experienced operator who has monitored their own patients over many years and has seen failed restorations they were responsible for placing, appreciated the causes of failure, and then repaired/replaced them. This individual can give an honest estimate of the longevity of the restorations placed by their own hand, incorporating into this the individual patient factors.

9.2.3 How may restoration outcome be assessed?

There are several indices available to clinical researchers to help to evaluate the causes and time lines for failure of restorations made from different dental materials (USPHS, Ryge & Snyder, and Hickel criteria to name but three; see QR code image 9.2). However, interpretation of collected, pooled data depends on what is designated a failure in the first instance. A restoration's success may be judged on:

- its clinical or radiographic appearance and technical form
- its clinical function
- whether the tooth is pain-/caries-free.

QR code 9.2 Scan this code with your mobile device to access the Hickel citeria.

www.ncbi.nlm.nih.gov/pubmed/20847997

When assessing patients in dental practice, either on primary examination or at the recall consultation, the aspects of a restoration to be assessed are shown in Table 9.1. Each of the criteria can be given scores on a numerical scale depending on the degree of 'failure.' Restoration failure is mechanical in origin. However, with experience and consideration of the clinical knowledge of the patient and the patient's views, the clinician is able to make a judgement as to the degree of failure and the necessity of operative intervention, in most cases. Note that a 'failed' restoration may require replacement in one patient, but a similar 'failure' in another patient may be accepted without operative intervention, depending on other factors. Therefore the decision as to whether to replace/repair the tooth–restoration complex will depend on input from both the experienced dentist and the patient.

9.2.4 How long should restorations last?

There are numerous factors that affect the answer to this important question that many patients will, quite reasonably, ask:

- The caries risk status of the patient (the higher the caries risk for a prolonged period of time, the less likely it is for restorations, independent of the restorative material used, to last as long without problems due to poor patient adherence to preventive regimes/maintenance of oral hygiene).
- The age of the patient. When clinical data from adolescents and adults are compared, it is found that restorations last longer in adults. This may reflect the susceptibility of younger people to caries, or differences in attitude/abilities with regard to dental care.
- The type and size/surface area of restorations. Smaller restorations with a reduced surface area exposed to the oral environment are easier to place and are easier for the patient to clean, as their margins will be more easily accessible to effective oral hygiene procedures and so will last longer than larger ones, if maintained.
- The restorative material used in the correct situation (see Section 9.2.2).
- The diagnostic criteria used by the dentist. This is particularly important with regard to recurrent/secondary caries (CARS), because this is the most common reason dentists give for replacing restorations (see later).
- The age and clinical experience of the operator. Young dentists, with less clinical experience, tend to replace more restorations than older dentists, as they have been exposed less to the deleterious consequences of the destructive operative restorative cycle on the natural life of teeth in their patients.
- Whether the dentist is reviewing their own work or that of another dentist. Changing dentists puts a patient 'at risk' of the diagnosis of failed restorations, and again is dependent on the criteria used to define failure, but this time without any prior background clinical information as to how and why the original restoration was placed.
- The care/attitude/motivation of the patient in maintaining their oral/dental health. Restorations are only ever as good as the operator placing them and the patient looking after them.

The actual answer to the question heading this section is 'It doesn't matter'! In the modern era of minimum intervention dental healthcare and minimally invasive operative treatments advocated throughout this text, the critical question to ask regarding the measure of a successful outcome of dental therapy/treatment is not necessarily about the technical quality and/or status of the restorations placed, but whether natural tooth structure and pulp sensibility have been preserved long term. It is the profession's duty to concern itself, along with the patient, with the maintenance of natural oral/dental health and pulp vitality as opposed to purely that of the restorations/prostheses placed to repair the consequences of dental disease. It is important to remember that

none of the man-made materials used to repair/replace natural tooth come close to replicating the qualities of natural tooth structure. These materials will always fail over varying periods of time, and will eventually be replaced by more recently developed products in materials technology, but these will still not be as good as biological tissues themselves. Therefore assessment of successful outcome should no longer be concerned with the simple overall longevity of restorations, but with the level of biological response that they elicit in the tooth and the ease with which they can be repaired when the time comes, without detriment to the remaining viable tooth. Thus this section should perhaps be titled 'How long should restored teeth last?'

9.3 Tooth failure

Teeth can fail for *mechanical/structural* reasons and/or *biological* reasons, either together with, or independently from, restoration failure (the tooth–restoration complex) (see Table 9.2).

Table 9.2 Causes of tooth failure

Tooth failure		Comments
Mechanical	Enamel margin	• Poor cavity design can leave weak, unsupported/undermined enamel margins which fracture under occlusal load • Cavity preparation techniques (burs) cause subsurface micro-cracks within the body of the enamel, so undermining the surface (see Chapter 5, Figure 5.34) • Adhesive shrinkage stresses on prisms at the enamel surface can cause them to be pulled apart, leading to cohesive marginal failure in the tooth structure, resulting in a micro-leakage risk (see Figure 9.12)
	Dentine margin	• Adhesive bond to hydrophilic dentine results in a poorer-quality bond (compared with that with sound enamel), which hydrolyses over time, leading to increased risk of micro-leakage • Deep proximal cavities often have exposed margins on dentine. Poor moisture control leads to compromised bonding technique, in turn increasing the risk of micro-leakage
	Bulk coronal/ cusp fracture	• Large restorations will weaken the coronal strength of the remaining hard tissue • Loss of marginal ridges/peripheral enamel will weaken the tooth crown • Cusps absorb oblique loading stresses and are prone to leverage/fracture (see Figure 9.13) • Can cause symptoms of food-packing, sensitivity
	Root fracture	• Often root-filled, heavily restored teeth (with post-core-crown) under heavy occlusal/lateral loads • Traumatic injury • Symptoms are variable (pain, mobility, tenderness on biting), but radiographic assessment is not always easy to interpret
Biological	Recurrent/ secondary/ caries/CARS	• New caries at a tooth–restoration gap with plaque biofilm accumulation and stagnation (see Figure 9.14) • Detected clinically or with radiographs (can be difficult to interpret) • Margin stain is not an indicator of recurrent caries (CARS) • Can affect a section of margin and not the whole restoration
	Pulp status	• Heavily restored teeth are more liable to pulp inflammation due to the original disease or the invasive direct operative treatment (see Figure 9.15) • Iatrogenic damage or ongoing disease may cause pulp pathology/necrosis
	Periodontal disease	• Examination of the periodontium is required for loss of attachment, pocket depths, and bone levels (see Figure 9.16) • Can be exacerbated by poor margin adaptation of restorations (causing plaque and debris stagnation)/margins encroaching into the periodontal biological width



Figure 9.12 A resin composite restoration in a maxillary premolar with a fine enamel crack evident on the palatal cusp (white line) remote to the tooth–restoration interface (arrow).

Q9.7: What has caused the enamel crack shown in Figure 9.12?



Figure 9.13 The mesiobuccal cusp of LR7 has fractured off due to excessive loading of the weakened crown and possible undermining of the cusp when the cavity was originally prepared.

Q9.8: Look at Figure 9.13. Which coronal structural feature is missing that has contributed to the increased weakness of the remaining tooth structure?

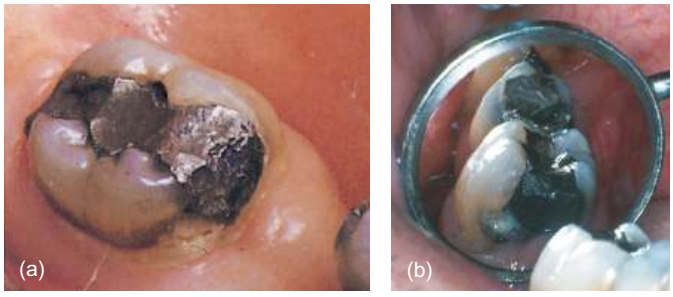


Figure 9.14 (**a**) Recurrent caries at the cervical margin gap between the tooth and amalgam in a mandibular molar. (**b**) Active, cavitated recurrent caries adjacent to an amalgam restoration with plaque retention.



Figure 9.15 Periapical radiograph of a heavily restored LL5; over time the pulp has become non-vital, and there is radiographic evidence of pulpal necrosis.

Q9.9: Look at Figure 9.15.

i Can you spot the evidence of pulpal necrosis?

ii What anatomical feature might confuse your diagnosis?

iii As the amalgam restoration has not extended into the pulp, why does it appear as though it has?

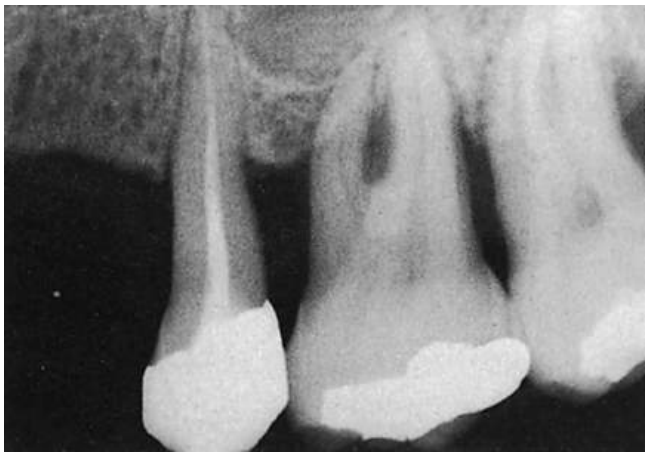


Figure 9.16 A periapical radiograph of a successful coronally restored UL6. Note the loss of alveolar bone support caused by periodontal disease leading to excessive mobility.

9.4 Monitoring the patient/course of the disease

9.4.1 Recall assessment and frequency

The recall visit follows a pattern similar to the initial assessment detailed in Chapter 2. The history will focus on what has happened since the dentist and patient last met. For instance, it is important to recheck the medical history carefully, but questions about the past dental history need not be asked again, except to check that no other dental treatment has been provided in the interim. When the clinical examination is carried out, particular attention is paid to areas noted as important or specifically requiring monitoring (e.g. caries, restorations, tooth wear) (see later).

9.4.2 Points to consider (especially for a previously high caries risk patient)

- Are existing restorations stable?
- Is there any clinical evidence for new lesions/demineralization?
- Is there evidence of lesion progression on radiographs (see Figure 9.17)?
- Are the dietary risk factors still present?
- Is the oral bacterial balance under control?
  - Check oral hygiene (procedures and disclosing solutions).
  - Consider repeating chair-side risk assessment tests (see Chapter 2)
- Has home care (topical remineralizing agents/high fluoride toothpastes, mouthwashes, etc.) helped? Has there been patient adherence to the documented non-operative preventive advice offered?

Visual and radiographic examination may be required, with repeated bitewing radiographs every 6 to 12 months for high risk/caries active individuals (see Figure 9.17). If the patient has modified the causative factors and reduced their risk, becoming caries inactive, then subsequent radiographs may only be required at 18- to 24-month intervals, but these are only approximate guidelines and will be subject to change depending on the patient's response to treatment and preventive advice. The intervals should be judged on an individual patient basis, just like the recall frequency (see Table 9.3).

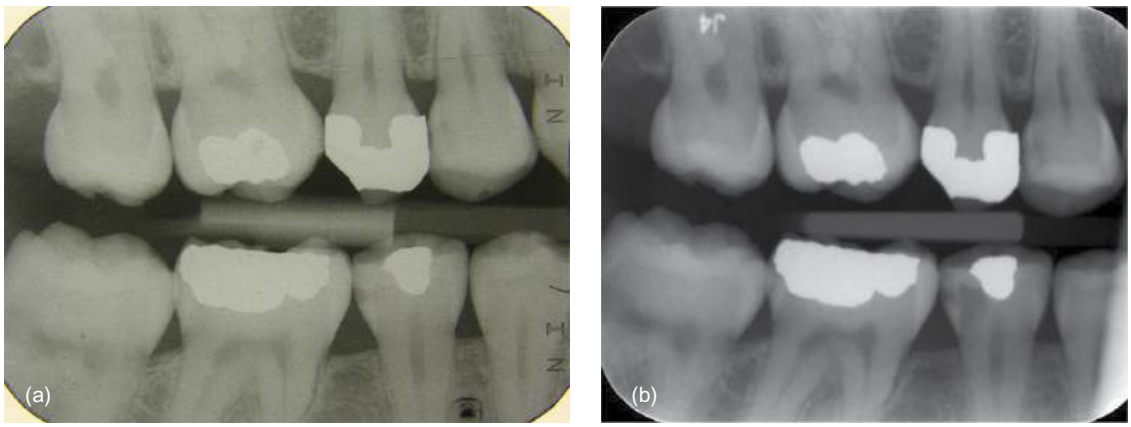


Figure 9.17 (**a**) Right bitewing radiograph of a high caries risk patient with early lesions evident distal UR4 and distal LR5. (**b**) Right bitewing radiograph of the same patient 11 months later showing significant progression of the two lesions. The caries control strategies discussed in Chapter 4 had not been fully implemented by either the dentist or the patient.

Table 9.3 Average recall frequency ranges for patients with respect to lesions present and level of caries susceptibility; note that recall frequencies must be tailored to the individual patient, and the figures presented are purely a guide (for more information about the mICDAS scores, see Table 2.3 in Chapter 2)

Identify	**Lesion**				
	Cavitated (mICDAS 3,4)	**Non-cavitated (mICDAS 1,2)**		**No lesion (mICDAS 0)**	
	High risk	*High risk*	*Low risk*	*High risk (unmodifiable factors)*	*Low risk*
Recall frequency	2–6 months	3–6 months	6–12 months	3–6 months	12–18 months

9.4.3 Monitoring tooth wear

As tooth wear is usually an ongoing problem on first presentation at the dental clinic, it is important to be able to monitor its progress in order to see if it is getting worse, and at what rate, or to see whether the controlling measures advised are having an effect on reducing the rate/stopping further progress. Monitoring methods may include the following:

- *Clinical digital photography*: standardized digital photography of the patient's teeth as shown in figures in this section, with good lighting, can be helpful for comparison over months and years. An overview can be gained from this rather than detailed measurements of any lesion change. Note that, in many countries, full written consent from the patient may be required before images can be captured and stored securely.
- *Tooth-wear indices*: these are clinical scoring systems that are available to permit objective numerical scores of the degree of tooth wear to be noted, reassessed, and compared at a later date. Some indices will also help with the care planning regimen required for the individual patient. Unfortunately, as with many of these indices, they are often complicated and time-consuming to use in a busy dental practice. Their use in research and epidemiological studies is recommended, but in general dental practice their use may be limited. However, practitioner-friendly tooth-wear screening indices do exist that help to give an overview of the patient's existing problem—for example, the Basic Erosive Wear Examination (BEWE) (see Table 2.4 in Chapter 2).
- *Serial, dated study models* (see Figure 9.18): Casts of the dentition can be made from impressions at 12- to 18-month intervals in patients where the tooth wear appears to be progressing, in order to be able to compare the changes and develop future care plans. Again, as with clinical photography, the changes will have to be quite extensive to be noticed. The patient should be given the dated casts to keep safely ensconced in bubble wrap and bring to future appointments. This may also help to evaluate better the attitude/motivation of the individual patient with regard to their tooth-wear problem. Remember that unless the patient appreciates their own responsibility for managing their condition, no form of treatment will be successful in the medium to long term.
- *Profilometry*: Using laboratory-based laser scanners, serial models can be scanned to give accurate changes in lesion dimensions over a period of time. This is a useful research tool at present. Intra-oral chair-side scanners, initially developed for fixed prosthodontic dental applications, are now being developed to measure this detailed level of tooth surface morphological change over time, and may prove useful in a dental practice setting.

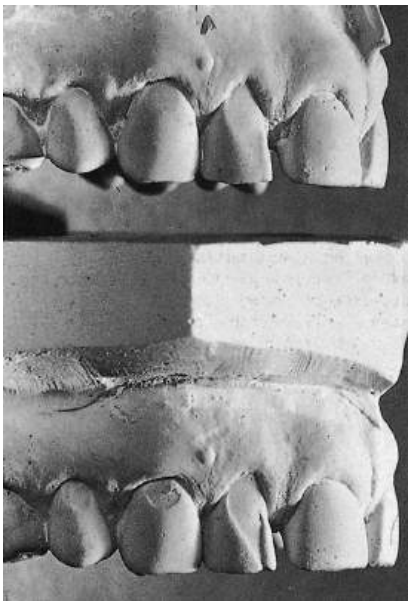


Figure 9.18 Serial study models taken with a 3-year interval showing the progress of tooth-wear lesions on the buccal aspects of UR234.

As with all cases where the progression of a relatively slowly developing lesion, disease, or condition is being measured, the relative impact of the perceived change on the patient's lifestyle plays a significant role in motivating the patient to alter their behaviour so as to prevent the occurrence of further disease. A patient is less likely to be concerned about a condition that is only developing slowly with little or no detrimental impact on their way of life. However, a rapid obvious deterioration in their health will have more impact, spurring them on to change their lifestyle. The screening index that has the most impact on the patient will require the appropriate level of explanation from members of the oral healthcare team in the clinic. Indices cannot be relied upon without this critical back-up.

9.5 Managing the failing tooth–restoration complex: the '5 Rs'

As was stated at the beginning of this chapter, the long-term maintenance and survival of the tooth–restoration complex are dependent upon regular and careful recall consultations to review not only the status of existing restorations, but also the patient's attitude to maintaining them with the appropriate level of well-executed preventive home care. Nevertheless, all tooth–restoration complexes have a finite lifespan, as has been discussed earlier. Modern restorations can be *refurbished, repaired*, and/or *replaced* when deemed necessary by the operator and the patient. The decision to repair or replace a restoration is one that has to be made specifically for a particular situation in a particular patient. When significant portions of the restoration (> 50%) have failed or the surrounding tooth structure is structurally and/or functionally compromised, partial or complete replacement of the restoration will be indicated. In most other cases, especially where the original restoration material type is known, careful minimally invasive repair of the deficient, fractured, or weakened sections may be appropriate. When removing old adhesive restorations completely, there is a significant chance of enlarging the cavity with a rotary instrument due to similarities in colour of the restoration and the surrounding tooth structure. Therefore other operative technologies may be useful, such as air abrasion (see Chapter

5, Section 5.8.4). Repairing failing portions of an existing restoration is the more conservative and minimally invasive option.

The minimally invasive non-operative/operative management options for the tooth–restoration complex that has been diagnosed as failing may be divided into five categories:

1. *Review* (see Figure 9.11): If only minor defects are evident, such as surface roughness/irregularities without concomitant plaque biofilm stagnation, the restoration can be monitored non-operatively. The primary factor in deciding to review is that by commencing operative treatment there will be no net clinical advantage gained by the patient. It is imperative that the patient is informed of the decision, with an explanation of its rationale, and that this is documented clearly in the notes, for both clinical and dento-legal purposes. Assessment of the patient's caries risk and their responsible adherence to non-operative preventive regimes is imperative. The use of intra-oral clinical photography is recommended to aid the reviewing of restorations long term. Care should be taken to try to standardize the perspective and lighting conditions between images taken. Suitable recall periodicity is required, and this will be dependent upon the attitude and interest that the patient has with regard to their own oral health.
2. *Refurbishment* (see Figure 9.19): This is indicated if there are small defects in the restoration which require intervention, including reshaping, removal of excess material, and/or polishing. If plaque stagnation occurs in such surface defects, and the patient cannot remove it easily with oral hygiene procedures, or there is an aesthetic benefit to the finished restoration, refurbishment is indicated. Micro-abrasion techniques to refresh resin composite surfaces, especially in the anterior aesthetic zone, would be included in this category.
3. *Resealing* (Figure 9.20): This is defined as the application of sealant into a non-carious, defective margin gap and/or surface. This is achieved through the use of flowable resin-based adhesive systems. Sealing a restoration minimally invasively prevents further ingress of plaque and the potential of caries developing.
4. *Repair*: This is the addition of a restorative material to an existing restoration. It may involve operative removal of the defective part of the restoration or modification of the existing retained portion to facilitate retention. A minimally invasive approach aims to restore the failing restoration while preserving the maximum quantity of tooth structure and maintaining pulp viability.
5. *Replacement*: This is the complete removal of the existing restoration and necessary tooth structure to prevent disease progression or to aid retention/stability. It is indicated if there are multiple or severe problems associated with the existing failed tooth–restoration complex, where a segmental repair is not feasible.

It must be appreciated that, in the majority of clinical cases, these categories of treatment are not mutually exclusive and cannot be considered separately. A more extensive surface refurbishment of a restoration together with resealing of its defective margins could be considered as a tooth–restoration complex repair. The categories have been defined to allow a better understanding of the minimally invasive approach to biological tooth preservation. The application of these procedures should not be considered a 'failure' of care, but rather part of the long-term management process of dental restorations. The '5 Rs' MI management protocol will be applied to different restorative materials in the following sections.

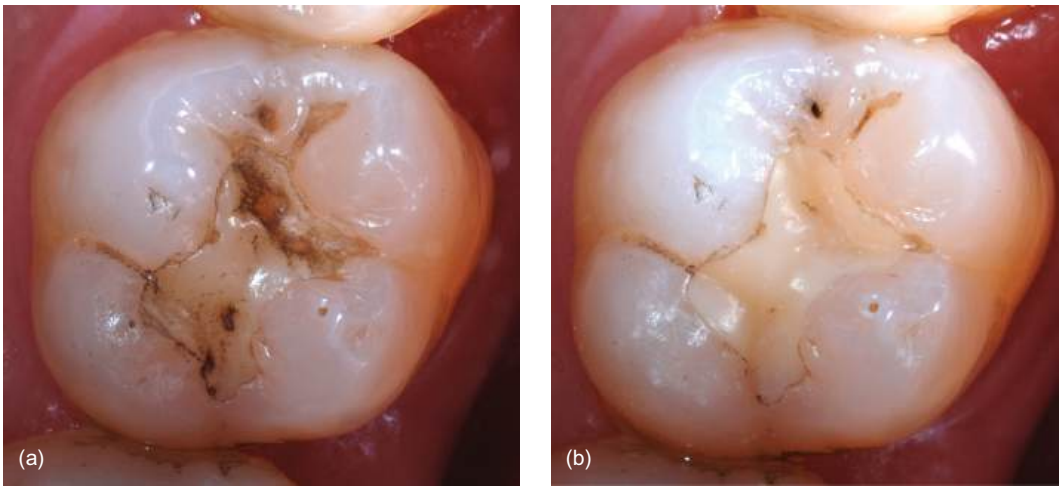


Figure 9.19 **(a)** A maxillary left molar with an old, stained, roughened occlusal resin composite Class I restoration. Especially in an at-risk patient with poorer oral hygiene, this restoration would encourage plaque stagnation. Many dentists would consider replacing a restoration with this appearance. **(b)** The same restoration after it has been refurbished simply using abrasive polishing discs/burs, taking only a few minutes. Its appearance has improved along with its morphology and smooth surface finish. This minimally invasive approach has indirectly increased the longevity of this restoration, as few dentists would now choose to replace it! (Courtesy of L Mackenzie.)

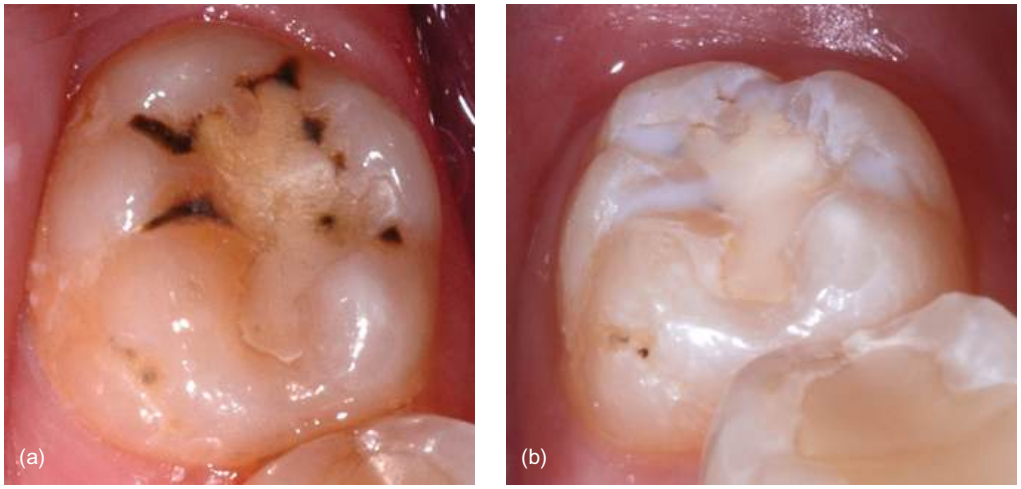


Figure 9.20 **(a)** Non-carious, stained, and defective margins associated with an occlusal resin composite restoration on a mandibular right second molar. **(b)** The defective margins were cleaned using air abrasion with 27-µm alumina particles, acid-etched for 20 seconds, wash and dried, adhesive placed, and resealed with a resin-based fissure sealant and photo-cured. (Courtesy of Dental Update.)

9.5.1 Dental amalgam

The most common causes of failure of dental amalgam restorations are secondary caries (pathological) and restoration/tooth fracture (mechanical) (see Tables 9.1 and 9.2).

- *Review*. If only minor deficiencies are present in the restoration, such as a minimal surface defect, reviewing the restoration periodically as per standard recall guidelines is a sensible management option. If there is no clinical advantage to treating the restoration operatively, it should be reviewed. Indeed, operative intervention of old, established amalgams can lead to significant levels of tooth tissue destruction, as the corroded alloy offers some 'support' to weakened tooth structure over time. Removing this can accelerate cusp /wall failure, thus complicating the replacement restoration significantly (see Figure 9.3a). Appropriate documentation and informed consent should be obtained from the patient.
- *Refurbishment or re-finishing* existing amalgam restorations (using brown and green amalgam abrasive rubber polishing points in a slow-speed handpiece; see Figure 5.31 in Chapter 5) is a useful treatment for anatomical form defects and for refreshing an old tarnished restoration. The refurbishment of anatomical form and surface roughness involves conservative and simple procedures which can enhance the appearance of amalgam restorations and, indirectly, their longevity, as they are less likely to be replaced by the dentist in the future.
- *Resealing*. The application of a resin composite sealant into a non-carious gap/defect at the margin of an amalgam restoration has been shown to increase the clinical longevity of the amalgam restoration (see Figure 9.21). Prior to application of a flowable resin composite, the amalgam surface and adjacent tooth can be modified using a variety of techniques. Air abrasion, etching with orthophosphoric acid, and application of a dentine bonding agent are all effective protocols cited in the literature.
- *Repair*. It is relatively easy to remove the remaining amalgam restoration/loose fragment. A tungsten carbide (TC) Beaver bur (see Figures 5.29 (a) and 5.33 (d) in Chapter 5) is used to cut through the amalgam from its centre towards the periphery of the portion to be removed (avoiding contact with the cavity margins). Large fragments are usually dislodged or can be flicked away using hand excavators (see Chapter 5).
  - Cavity margins may be 'freshened up' using rotary instrumentation to remove stain or early demineralized hard tissues, especially if the restoration is to be repaired/replaced using a tooth-coloured adhesive restorative material.
  - Retention has to be gained macro-mechanically via cavity undercuts, slots, or grooves. If replacing the complete restoration, these retentive features will often already feature in the existing cavity design and may only need slight modification. When repairing the damaged portion of the restoration/tooth only, these retentive design features should be cut into the retained portion of the original amalgam restoration to avoid further loss and weakening of the surrounding tooth structure (see Figure 9.22a).
  - Some may advocate the use of chemical retention, using a bonded amalgam technique, in an attempt to conserve the remaining tooth structure, but there is limited evidence for the long-term success rates of this repair technique (see Chapter 7, Section 7.5.3, and Chapter 8, Section 8.9).
  - If the previous steps can be carried out, the repair can be completed using amalgam (see Figure 9.22b). If not, resin composite (including the use of an etch-and-rinse adhesive, types 1 and 2) or chemically adhesive GIC in situations with poor moisture control might be the materials of choice. The margins must be contoured appropriately, avoiding ledges, overhangs, and voids.

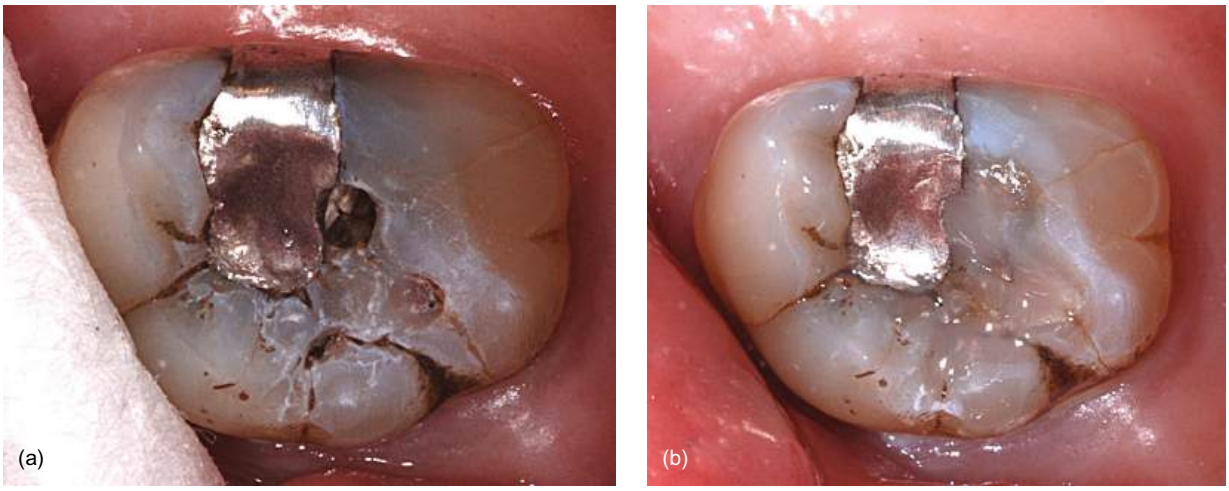


Figure 9.21 **(a)** The occlusal surface of a UR7 with a Class II mesio-occlusal (MO) amalgam restoration. Note the non-carious defect associated with the palatal margin of the amalgam. **(b)** The defect has been cleaned, acid-etched, adhesive applied, and resealed using flowable resin composite. (Courtesy of L. Mackenzie.)

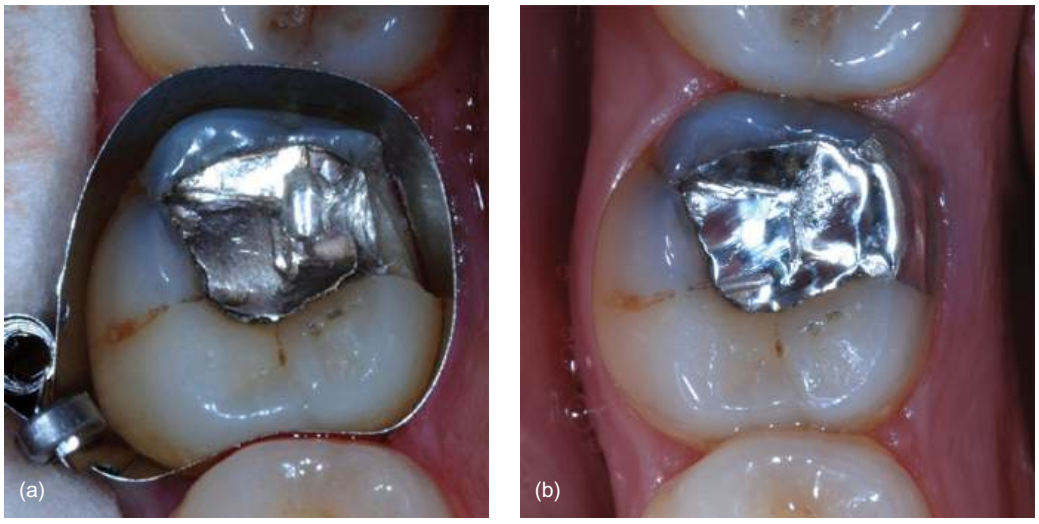


Figure 9.22 **(a)** LR6 with an old occlusal amalgam and a recently fractured disto-lingual cusp. A retentive lock has been cut into what was originally an occlusal amalgam to improve retention for the repair. **(b)** The repair of the tooth–restoration complex has been successful using amalgam. The image was taken 2 years after the repair was originally carried out. (Courtesy of Dental Update.)

Q9.10: Look at Figure 9.22 (a). How else can the amalgam repair bond to the old restoration be enhanced?

9.5.2 Resin composites/GIC

In contrast to dental amalgam, resin composite restorations present a more complicated chemical substrate for repair. There is variation in the chemical composition of resin composites with regard to the type of resin matrix, with additional disparities in inorganic filler quantity and type. The age of the restoration also has a significant deleterious effect on the bond strength between new and old resin composite repairs. Newly placed resin composites, placed incrementally, have cohesion between increments due to the presence of uncured resin monomer after photo-curing—the oxygen-inhibited layer. This degrades rapidly in clinical service and reduces the success of bonding further additions of resin composite for intra-oral repairs.

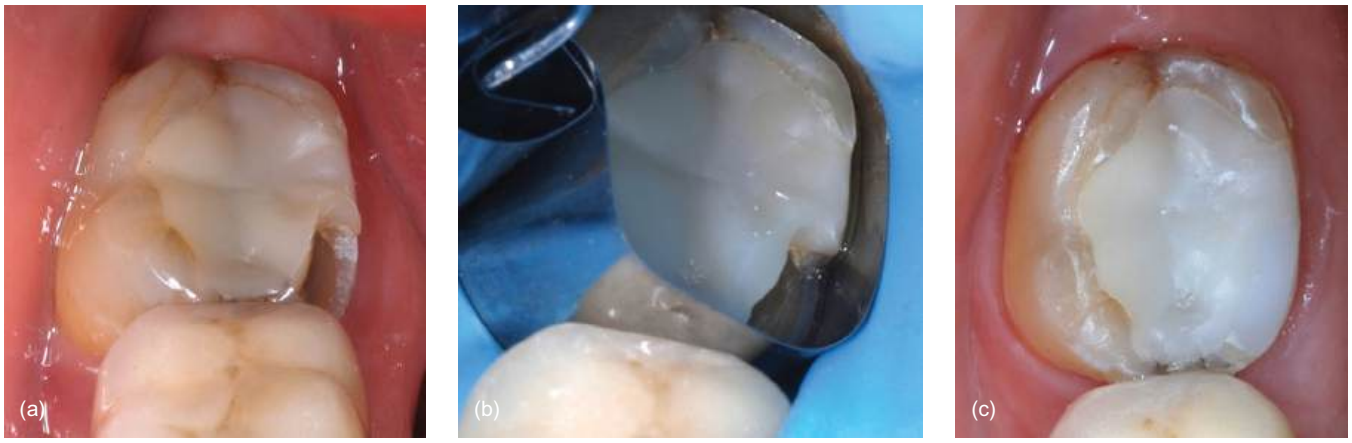


Figure 9.23 **(a)** LR7 with an occluso-lingual resin composite restoration and a fractured mesio-lingual cusp. **(b)** The isolated cavity walls can be freshened with rotary instruments or air abrasion. The cavity is then silanated to improve the resin composite–composite bond, and adhesive applied and photo-cured. **(c)** The immediate post-operative image of the resin composite–composite repair. (Courtesy of Dental Update.)

Q9.11: What type of matrix system has been used in the restoration shown in Figure 9.23 (b)?

Dentists have been shown to be more likely to intervene operatively with a resin composite restoration than with an amalgam restoration. It has been speculated that this difference may be due to the perceived increased longevity of amalgam restorations in general, and the lack of correlation between margin breakdown and the presence of caries.

- It is more usual to refurbish, reseal, or repair adhesive restorations, rather than replace them completely. Polishing, improvement in surface roughness, and anatomical form of Class I and II resin composite restorations have been shown to maintain an improvement for 3 years post intervention (see Figure 9.19). If there is a gross aesthetic or biological concern, old restorations may be veneered/re-surfaced with new material (the old restoration being resurfaced with an up-to-date, shade-matched equivalent, rather than replaced completely).
- Repair. Use rotary instrumentation to remove the defective portion of restoration. Care must be taken to avoid cavity over-preparation, as it can be difficult to distinguish between the adhesive restoration and the cavity margin due to colour similarities. Air-abrasive techniques using alumina or bioactive glass powders might facilitate this process, due to their inherent selectivity for resin composites and/or grossly demineralized enamel.
- Fresh cavity margins should expose a roughened enamel and/or dentine surface for better micro-mechanical/chemical adhesion.
  - *GIC repair*: condition all surfaces with 10% polyacrylic acid for 10 seconds, wash thoroughly, blot dry, and apply/pack GIC (with matrixing as appropriate).
  - *Resin composite repair* (see Figure 9.23): acid-etch with 37% orthophosphoric acid for 20 seconds (enamel) and 10–15 seconds (dentine), wash thoroughly (10 seconds), and dry.
  - Silanating agent may be used to couple new methacrylate-based resin composite to old (painted on the bonding surface of the old restoration and evaporated). This step may be omitted, as it can interfere with the adhesive chemistry of the specific dental adhesive used, and also it may be clinically difficult to separate these two procedures.

    Dentine bonding agent is agitated on to cavity surfaces, gently dried (until no rippling is noticed, indicating solvent evaporation), and light-cured.

    Resin composite is added in small increments, photo-cured, and finished appropriately (see Figure 5.32 in Chapter 5).
- Another factor in the success of resin composite repairs is appreciating the chemistry of the original material and the ability to match new material with old. The use of a dental adhesive also significantly improves outcomes, mainly by increasing the 'wetting' or adaptation of the repairing material.

9.6 Answers to self-test questions

Q9.1: What restorative material should be used to replace the amalgam shown in Figure 9.1?

A: Dental resin composite.

Q9.2: What would be the difficulty faced by the dentist when replacing the Class III restoration shown in Figure 9.2?

A: Aesthetics—matching the natural translucency from the incisal edge through to the mesial aspect of the UL2. An aesthetic layered composite restoration would have to be placed carefully mimicking the underlying dentine and overlying enamel shades.

Q9.3: Look at Figure 9.5. What has caused this positive ledge to occur?

A: A poorly adapted/wedged matrix band at the cervical margin of the approximal Class II cavity.

Q9.4: Look at Figure 9.6. What periodontal condition has been caused by the accumulated plaque, and what is its relevance when replacing the restorations?

A: Chronic marginal gingivitis. This would need to be remedied with improved oral hygiene techniques prior to any restoration being placed, in order to ensure optimal moisture control during placement of the new composite restorations.

Q9.5: What factors have resulted in the fracture shown in Figure 9.8?

A: Poor cavity design—the mesial portion of the cavity was too shallow and the amalgam fractured under occlusal load due to its inherent weakness in thin section and lack of macro-mechanical retentive features. Inappropriate choice of material—a resin composite might have been a better choice, thereby preventing any unnecessary extension of the cavity.

Q9.6: Look at Figure 9.10. Does this finding necessitate the replacement of this restoration?

A: No. The restoration is fully functional and is not causing any clinical problems.

Q9.7: What has caused the enamel crack shown in Figure 9.12?

A: It has been caused by shrinkage stress from the composite pulling on the prismatic enamel structure and causing cohesive failure in the enamel.

Q9.8: Look at Figure 9.13. Which coronal structural feature is missing that has contributed to the increased weakness of the remaining tooth structure?

A: A missing mesial marginal ridge.

Q9.9: Look at Figure 9.15.

i Can you spot the evidence of pulpal necrosis?

A: Widening of the periodontal ligament space at the root apex, with loss of lamina dura in this area of the LL5.

ii What anatomical feature might confuse your diagnosis?

A: An overlying mental foramen.

iii As the amalgam restoration has not extended into the pulp, why does it appear as though it has?

A: There is an additional buccal cervical amalgam restoration, the radiographic opacity of which has superimposed itself over the pulp chamber radiolucency.

Q9.10: Look at Figure 9.22 (a). How else can the amalgam repair bond to the old restoration be enhanced?

A: By using air abrasion/burs to roughen the surfaces, and then using a resin cement in a bonded amalgam technique. However, there is inconclusive evidence as to whether this actually strengthens the adhesive bond between the two amalgams.

Q9.11: What type of matrix system has been used in the restoration shown in Figure 9.23 (b)?

A: A circumferential metal matrix band. This is required because a sectional band would not extend to help to contour the mesio-lingual aspect of the restoration.